IoT Machinery Vibration Monitoring & Alarm System

Hardware Assembly Manual & Wiring Specification

Project Scope: The objective of this project is to construct a localized, hardware-driven diagnostic system that detects structural shifts or machinery faults through physical vibration. Assembled on a standard breadboard, the system integrates an ESP32-WROOM-32 microcontroller as the primary processor, an SW-420 digital vibration sensor to track environmental tremors, and a passive buzzer to provide immediate local acoustic feedback. To protect the microprocessing unit from overcurrent damage, a BC547 NPN transistor is configured as an electronic switch to isolate the load, paired with an inline 1kΩ base resistor for signal protection.

Hardware Master Wiring Reference

Component	Component Pin	Connects to ESP32 / Destination	Notes
ESP32 Board	GND Pin	Breadboard GND Rail (-)	Establishes common system ground.
ESP32 Board	3V3 Pin	Breadboard Power Rail (+)	Supplies 3.3V power to components.
SW-420 Sensor	VCC	Breadboard Power Rail (+)	Receives operational voltage.
SW-420 Sensor	GND	Breadboard GND Rail (-)	Sensor ground path.
SW-420 Sensor	DO (Digital Out)	GPIO 18	Transmits the vibration signal state.
BC547 Transistor	Collector (Left Pin)	Buzzer Negative (-)	Acts as the switch terminal for the buzzer.
BC547 Transistor	Base (Middle Pin)	GPIO 23 (via 1kΩ Resistor)	Controls switch state; resistor prevents overcurrent.
BC547 Transistor	Emitter (Right Pin)	Breadboard GND Rail (-)	Drains switched buzzer current to ground.
Passive Buzzer	Positive (+)	Breadboard Power Rail (+)	Receives constant system power.
Passive Buzzer	Negative (-)	Transistor Collector	Connected to the switching node.

Step-by-Step Hardware Assembly

Step 1 Establish the Power Rails & Mount Core ICs

- **Insert the ESP32:** Press your ESP32-WROOM-32 module into the center of the breadboard so it straddles the middle divider ditch. Make sure its pins fit firmly into separate row holes.
- **Wire the Main Rails:** Run a jumper wire from the ESP32 3V3 pin to the red positive (+) outer rail of the breadboard. Run a second jumper wire from an ESP32 GND pin to the black negative (-) outer rail.
- **Mount the Transistor:** Orient the BC547 Transistor so its flat face is pointing toward you. Push its three legs into three adjacent, empty holes on a single row of the breadboard.
- **Pin Check:** From left to right, the pins are Collector, Base, and Emitter.

Step 2 Wire the SW-420 Vibration Sensor

- **Mount the Sensor Module:** Place the SW-420 sensor module onto an open section of the breadboard.
- **Connect Power:** Use a jumper wire to connect the sensor's VCC pin to the red positive (+) rail. Connect its GND pin to the black negative (-) rail.
- **Connect Data:** Route a jumper wire from the sensor's DO (Digital Out) pin to the hole row inline with the ESP32's GPIO 18 pin.

Step 3 Integrate the Transistor Switching Circuit & Buzzer

- **Place the Protection Resistor:** Take your 1kΩ Resistor. Insert one end into the row containing the ESP32's GPIO 23 pin. Insert the other end directly into the row containing the BC547's center leg (Base). *Do not skip this step, or you risk damaging the ESP32 pin.*
- **Ground the Transistor Switch:** Run a jumper wire from the BC547's right leg (Emitter) straight to the black negative (-) rail.
- **Hook up the Buzzer:** Insert your Passive Buzzer into the board. Connect the longer pin (Positive +) to the red positive (+) power rail. Connect the shorter pin (Negative -) to the row containing the BC547's left leg (Collector).

Expanded Remote Notification Options

Here is the complete menu of internet communication methods for your ESP32 vibration sensor, categorized by how you receive the data:

Category 1: Standard Browser Messaging (No Specialized Apps Required)

These options run directly inside your existing web browser profiles or built-in phone features.

• Telegram Web Interface (Recommended App-Free Pick)

- **The Workflow:** The ESP32 pings the secure Telegram API Engine using an HTTPS link.
- **The Access Point:** You keep a standard browser tab pinned to Telegram Web. When vibrations trigger, the server drops an instant chat notification right onto your active browser tab screen.

• WhatsApp Web / CallMeBot Gateway

- **The Workflow:** The ESP32 sends a basic web request to the public CallMeBot Messaging Hub.
- **The Access Point:** The gateway processes the link and forwards a real-time text message straight to your personal WhatsApp profile. You can view this completely app-free using WhatsApp Web in any standard mobile or desktop browser.

• Direct-to-SMS Texting (Twilio Bridge)

- **The Workflow:** The ESP32 executes an API call out to the Twilio SMS Platform.
- **The Access Point:** Twilio converts the network packet into a standard cellular network text message. It delivers straight to your phone's native, built-in SMS inbox without requiring any internet data or messaging apps to be open.

Category 2: Native System & Local Display Upgrades

These options provide dedicated control interfaces without relying on standard mobile notification wrappers.

• Local Browser Warning Dashboard

- **The Workflow:** The ESP32 completely hosts its own local web server directly on your home Wi-Fi network routing path.
- **The Access Point:** You open your normal browser (like Chrome or Safari) and go to the ESP32's local IP address. If the machine shakes, a background JavaScript loop instantly transforms the view into a bright red flashing screen and forces your browser to play a loud warning siren sound.

• Automated Email Routing (SMTP Server)

- **The Workflow:** The ESP32 logs into an encrypted email server using an SMTP helper script.
- **The Access Point:** It fires a high-priority warning message directly to your existing email inbox. You receive it instantly through whatever standard email tool or website you already use.